Linear Algebra Recitation Class: Week 1

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Through out this notes, $\mathbb{Q}$ is the set of all rational numbers, $\mathbb{R}$ is the set of all real numbers, and $\mathbb{C}$ is the set of all complex numbers.

Definition 1. Let F be $\mathbb{Q}$ or $\mathbb{R}$ or $\mathbb{C}$. A **vector space** over F (or an F vector space) is a nonempty set V together with two operations:

$$\begin{aligned} \text{(vector addition)} : V \times V &\longrightarrow V \\ (v_1, v_2) &\longmapsto v_1 + v_2 \end{aligned} \qquad \begin{aligned} \text{(scalar multiplication)} : F \times V &\longrightarrow V \\ (a, v) &\longmapsto a \cdot v \end{aligned}$$

satisfying:

VS1. For any $v_1, v_2 \in V$, $v_1 + v_2 = v_2 + v_1$.

VS2. For any $v_1, v_2, v_3 \in V$, $v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$.

VS3. There exists an element $0_V \in V$ such that $v + 0_V = v = 0_V + v$ for any $v \in V$.

VS4. For any $v \in V$, there exists an element $v' \in V$ such that $v + v' = 0_V$.

VS5. For any $v \in V$, $1 \cdot v = v$. (Note that $1 \in F$.)

VS6. For any $a, b \in F$, $v \in V$, $(a \cdot_F b) \cdot v = a \cdot (b \cdot v)$.

VS7. For any $a \in F$, $v_1, v_2 \in V$, $a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$.

VS8. For any $a, b \in F$, $v \in V$, $(a +_F b) \cdot v = (a \cdot v) + (b \cdot v)$.

where $\cdot_F, +_F$ are the multiplication operator and the addition operator in F.

Example. The vector addition and the scalar multiplication of a vector space need NOT be those we normally use. For example, consider $\mathbb{R}^+$ be the set of all positive real number, i.e.

$$\mathbb{R}^+ := \{r \in \mathbb{R} \mid r > 0\}.$$

For any $v_1, v_2, v \in \mathbb{R}^+$, $a \in \mathbb{Q}$, we define

$$v_1 \boxplus v_2 := v_1 v_2 \text{ (standard multiplication)} \qquad q \boxtimes v := v^q \text{ (standard exponentiation)}.$$

Then $\mathbb{R}^+$ is a vector space over $\mathbb{Q}$. Let's check this directly:

VS1. Given any $v_1, v_2 \in \mathbb{R}^+$, $v_1 \boxplus v_2 = v_1 v_2 = v_2 v_1 = v_2 \boxplus v_1$. ✓

VS2. Given any $v_1, v_2, v_3 \in \mathbb{R}^+$,

$$v_1 \boxplus (v_2 \boxplus v_3) = v_1 \boxplus (v_2 v_3) = v_1 v_2 v_3 = (v_1 v_2) \boxplus v_3 = (v_1 \boxplus v_2) \boxplus v_3. \checkmark$$

VS3. Define $0_{\mathbb{R}^+} := 1 \in \mathbb{R}^+$, then given $v \in V$, we have

$$v \boxplus 0_{\mathbb{R}^+} = v \cdot 1 = v = 1 \cdot v = 0_{\mathbb{R}^+} \boxplus v. \checkmark$$

VS4. Given any $v \in \mathbb{R}^+$, we define $v' := 1/v \in \mathbb{R}^+$, then

$$v \boxplus v' = v \cdot 1/v = 1 = 0_{\mathbb{R}^+}. \checkmark$$

VS5. Given any $v \in \mathbb{R}^+$, $1 \boxtimes v = v^1 = v$. ✓ (Note that the 1 here is the 1 in $\mathbb{Q}$)

VS6. Given any $a, b \in \mathbb{Q}$, $v \in \mathbb{R}^+$,

$$(a \cdot_{\mathbb{Q}} b) \boxtimes v = v^{ab} = v^{ba} = (v^b)^a = a \boxtimes v^b = a \boxtimes (b \boxtimes v). \checkmark$$

VS7. Given any $a \in \mathbb{Q}$, $v_1, v_2 \in \mathbb{R}^+$,

$$a \boxtimes (v_1 \boxplus v_2) = a \boxtimes (v_1 v_2) = (v_1 v_2)^a = v_1^a v_2^a = v_1^a \boxplus v_2^a = a \boxtimes v_1 + a \boxtimes v_2. \checkmark$$

VS8. Given any $a, b \in \mathbb{Q}$, $v \in \mathbb{R}^+$,

$$(a +_{\mathbb{Q}} b) \boxtimes v = v^{a+b} = v^a v^b = v^a \boxplus v^b = (a \boxtimes v) \boxplus (b \boxtimes v). \checkmark$$

We introduce some algebraic structures other than vector spaces, most of which will be learned in Abstract Algebra courses.

Definition 2. A **group** is a nonempty set G with a binary operation

$$\begin{aligned} \cdot : G \times G &\longrightarrow G \\ (g_1, g_2) &\longmapsto g_1 \cdot g_2 \end{aligned}$$

satisfying:

G1. For any $g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.

G2. There exists an element $e \in G$ such that $g \cdot e = g = e \cdot g$ for any $g \in G$.

G3. For any $g \in G$, there exists $g' \in G$ such that $g \cdot g' = e = g' \cdot g$.

Furthermore, if the condition

Ga. For any $g_1, g_2 \in G$, $g_1 \cdot g_2 = g_2 \cdot g_1$.

is satisfied. We say G is **abelian** (or G is an abelian group), named after a Norwegian mathematician Niels Henrik Abel.

Remark. If G is a group, it can be proved that the element e defined in G2 is unique, which is called the **identity** of G. Given any element $g \in G$, the element g' in G3 is also unique, which is called the **inverse** of g.

Example. The set of integer $\mathbb{Z}$ with the standard addition $+$ is an abelian group. The identity element of $\mathbb{Z}$ is 0 and given $z \in \mathbb{Z}$, the inverse of z is $-z$.

Example (General linear groups)**.** The set of all 2×2 invertible matrix with coefficients in $\mathbb{R}$

$$\mathrm{GL}_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : ad - bc = 0, \quad a, b, c, d \in \mathbb{R} \right\}$$

with matrix multiplication forms a group. It is NOT abelian. For example,

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}.$$

Example. If V is a vector space over F, note that [VS1] and [Ga] are equivalent conditions, similarly [VS2], [G1]; [VS3], [G2]; and [VS4], [G3] are pairs of equivalent conditions. Therefore, a vector space over F with vector addition forms an abelian group.

Exercise A. Think one example of abelian group and one for non-abelian group other than those mentioned above.

Definition 3. A **ring** is a nonempty set R with two binary operations

$$\begin{aligned} + : R \times R &\longrightarrow R \\ (r_1, r_2) &\longmapsto r_1 + r_2 \end{aligned} \qquad \begin{aligned} \cdot : R \times R &\longrightarrow R \\ (r_1, r_2) &\longmapsto r_1 \cdot r_2 \end{aligned}$$

satisfying

- Ra. R is an abelian group with $+$, the identity element is denoted by 0_R
- R1. For any $r_1, r_2, r_3 \in R$, $r_1 \cdot (r_2 \cdot r_3) = (r_1 \cdot r_2) \cdot r_3$.
- R2. There exists an element $1_R \in R$, such that $r \cdot 1_R = r = 1_R \cdot r$ for any $r \in R$.
- R3. For any $r_1, r_2, r_3 \in R$, $r_1 \cdot (r_2 + r_3) = r_1 \cdot r_2 + r_1 \cdot r_3$.
- R4. For any $r_1, r_2, r_3 \in R$, $(r_1 + r_2) \cdot r_3 = (r_1 \cdot r_3) + (r_2 \cdot r_3)$.

If R furthermore satisfies

- Rc. For any $r_1, r_2 \in R$, $r_1 \cdot r_2 = r_2 \cdot r_1$.

Then we say R is **commutative**, or R is a commutative ring.

Remark. Similar as groups, if R is a ring, then 0_R, 1_R is unique, called the **additive identity** and **multiplicative identity** of R. Given any element $r \in R$ and see R as a group with $+$, the unique inverse of r is called the **additive inverse** of r, denoted by $-r$.

Example. $\mathbb{Z}$ is a commutative ring equipped with standard addition and multiplication, where $0_{\mathbb{Z}} = 0$, $1_{\mathbb{Z}} = 1$. Note that the product of any two nonzero integer is nonzero, which gives us a motivation to define the following algebraic structure.

Definition 4. A commutative ring R is called an **integral domain** if $0_R \neq 1_R$ and the product of any two elements that are not 0_R is not 0_R.

Example (Matrix rings)**.** The set of all 2×2 matrix with coefficients in $\mathbb{R}$

$$R := M_2(\mathbb{R}) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}$$

with matrix addition and matrix multiplication forms a ring, where 0_R is the zero matrix, and 1_R is the identity matrix. It is NOT commutative (see the counterexample of $\mathrm{GL}_2(\mathbb{R})$ above).

Remark. We also say that $M_2(\mathbb{R})$ is an $\mathbb{R}$**-algebra**, which is, roughly speaking, a ring and a vector space over $\mathbb{R}$ at the same time with some compatibility with scalars. (If you are interested, you can look the definition up in `https://en.wikipedia.org/wiki/Algebra_over_a_field`, and replace every field K by $\mathbb{R}$ if needed.)

Example (Polynomial rings)**.** The set of polynomial with coefficients in $\mathbb{R}$

$$\mathbb{R}[x] := \{f(x) \mid f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad a_1, \cdots, a_n \in \mathbb{R}, n \geq 0\}$$

forms a commutative ring with standard polynomial addition and polynomial multiplication. The additive identity of $\mathbb{R}[x]$ is the zero polynomial 0 and the multiplicative identity is the constant function 1.

Remark. Similar as above, $\mathbb{R}[x]$ is a commutative $\mathbb{R}$-algebra, i.e. an $\mathbb{R}$-algebra which is also a commutative ring.

Exercise B. Think one example of commutative ring and one for non-commutative ring other than those mentioned above.

One can see that $\mathbb{Z}$ and $\mathbb{Q}$ are commutative rings with the standard addition and multiplication. While for any nonzero element $q = a/b \in \mathbb{Q}$, we can always find an element $q' = b/a \in \mathbb{Q}$, called the (because it is unique) **multiplicative inverse** of q, such that $qq' = 1$, we cannot do the same thing with every nonzero element in $\mathbb{Z}$, since $\mathbb{Z}$ does not contain some fractions. This gives the very difference between a commutative ring and a field.

Definition 5. A **field** F is a commutative ring with $1_F \neq 0_F$ where every nonzero element has its multiplicative inverse.

Remark. If we do not require F to be commutative, then F is called a **division ring** or a **skew field**.

Exercise C. Check that $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are all fields. Prove that

$$\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

is a field. (Hint: What is the multiplicative inverse of $a + b\sqrt{2}$ when $a, b \neq 0$?)

Exercise D. If F, K are fields and $F \subseteq K$, we say F is a **subfield** of K. Prove that if F is a subfield of K, then K is a vector space over F. This is a basic yet really important fact in field theory.

Lastly, we introduce an algebraic structure similar to a vector space over a field.

Definition 6. Let R be a ring, a **left R-module** or a module over R is a nonempty set M together with two operations:

$$\begin{aligned} + : M \times M &\longrightarrow M & \cdot : R \times M &\longrightarrow M \\ (m_1, m_2) &\longmapsto m_1 + m_2 & (a, m) &\longmapsto a \cdot m \end{aligned}$$

satisfying:

M1. For any $m_1, m_2 \in M$, $m_1 + m_2 = m_2 + m_1$.

M2. For any $m_1, m_2, m_3 \in M$, $m_1 + (m_2 + m_3) = (m_1 + m_2) + m_3$.

M3. There exists an element $0_m \in M$ such that $m + 0_M = m = 0_M + m$ for any $m \in M$.

M4. For any $m \in M$, there exists an element $m' \in M$ such that $m + m' = 0_M$.

M5. For any $m \in M$, $1_R \cdot m = m$.

M6. For any $a, b \in R$, $m \in M$, $(a \cdot_R b) \cdot m = a \cdot (b \cdot m)$.

M7. For any $a \in R$, $m_1, m_2 \in M$, $a \cdot (m_1 + m_2) = a \cdot m_1 + a \cdot m_2$.

M8. For any $a, b \in R$, $m \in V$, $(a +_R b) \cdot m = (a \cdot m) + (b \cdot m)$.

where $\cdot_R, +_R$ are the multiplication operator and the addition operator in R.

Remark. One can also define a right R-module with some modification of the definition above. When R is commutative, the notion of left and right R-module coincide. Since the axioms [VS*] of vector spaces are just the counterpart of [M*], one can think an R-module is some kind of "vector space" over a ring.

Exercise E. If F is a field (which is also a ring by definition), convince yourself that a left F-module is a right F-module and a vector space over F.

Example. Any abelian group G with binary operation $+$ can be seen as a $\mathbb{Z}$-module via.

$$\begin{aligned} + : G \times G &\longrightarrow G & \cdot : \mathbb{Z} \times G &\longrightarrow G \\ (g_1, g_2) &\longmapsto g_1 + g_2 & (n, g) &\longmapsto n \cdot g = \begin{cases} \underbrace{g + g + \cdots + g}_{n \text{ times}} & \text{if } n \geq 0 \\ \text{inverse of } |n| \cdot g & \text{if } n < 0 \end{cases} \end{aligned}$$

Example. Let $M = \mathbb{R}^2 = \begin{pmatrix} a \\ b \end{pmatrix}$, $R = M_2(\mathbb{R})$, then M is a left R-module with addition and multiplication define by

$$\begin{aligned} + : M \times M &\longrightarrow M & \cdot : R \times M &\longrightarrow M \\ \left(\begin{pmatrix} a_1 \\ b_1 \end{pmatrix}, \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} \right) &\longmapsto \begin{pmatrix} a_1 + a_2 \\ b_1 + b_2 \end{pmatrix} & \left(\begin{pmatrix} s & t \\ u & v \end{pmatrix}, \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} \right) &\longmapsto \begin{pmatrix} s & t \\ u & v \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} \end{aligned}$$

Exercise F. Check that (or convince yourself) that these two examples have module structures.

References

[FIS03] Stephen H. Friedberg, Arnold J. Insel, and Lawrence E. Spence. *Linear algebra.* Prentice Hall, Inc., Upper Saddle River, NJ, fourth edition, 2003.

[Fra02] John B. Fraleigh. *A first course in abstract algebra.* Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 2002.